Predictive Analysis of US Flight Delays

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Abstract

Flight delays are a common issue that impact both passengers and airline operations. This project uses historical U.S. flight data to develop predictive models that estimate whether a flight will be delayed. Using a Random Forest model and exploratory data analysis, key patterns and factors influencing delays were identified. Results show that while the model performs well overall, challenges remain in accurately predicting less frequent delay events.

Introduction

- Flight delays affect:
  - Passenger travel plans
  - Airline costs and scheduling
- Airlines rely on data to reduce delays
- Understanding delay patterns can improve efficiency
- This project focuses on predicting delays using historical data

Objectives

- Predict whether a flight will be delayed
- Identify key factors influencing delays
- Compare performance of multiple models
- Analyze patterns in U.S. flight data

Methods, Materials, and Models

Dataset
- Source: Bureau of Transportation Statistics
- Timeframe: December 2024
- ~630,000 flight records

Features
- Airline carrier
- Origin and destination airports
- Scheduled and actual departure times
- Delay and cancellation status

Tools
- Python
- Pandas, NumPy
- Scikit-learn
- Matplotlib

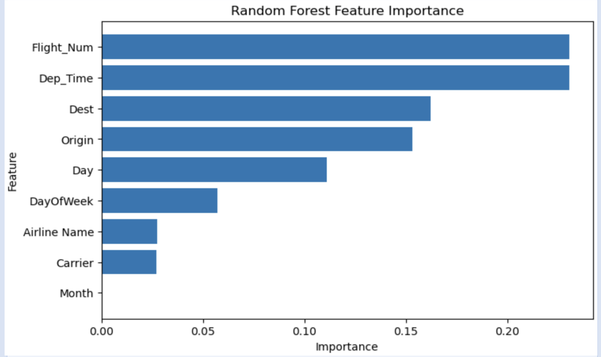
Models
- Logistic Regression (baseline)
- Random Forest (primary model)

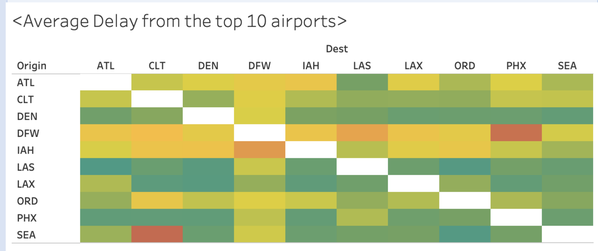
Results

The Random Forest model achieved an overall accuracy of approximately 80%, performing well in predicting non-delayed flights. However, it was less effective at identifying delayed flights, likely due to class imbalance in the dataset.

Analysis showed that many delayed flights were misclassified, indicating the need for further model tuning. Feature importance results identified departure time, origin, destination, and flight number as the most influential predictors.

Overall, the model captures general delay patterns but requires improvement for detecting less frequent delay events.

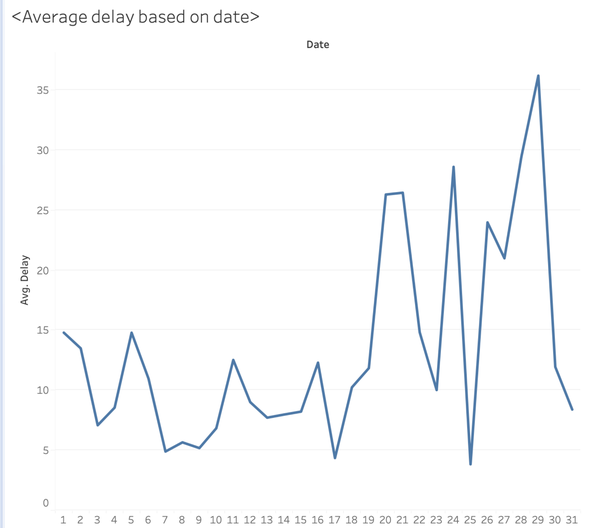




Conclusion and Future Work

- Model successfully identifies general delay patterns
- Performance is limited by class imbalance
- Predicting delays remains a complex problem

Future Work
- Explore dates/holidays
- Improve class balance techniques
- Add additional variables
- Test additional machine learning models
- Incorporate real-time data



References

- Bureau of Transportation Statistics

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